# AI5707 – PATTERN RECOGNITION AND MACHINE LEARNING

ASSIGNMENT – 8

Normalsize LDA, KNN, SVM, GMM and Dynamic Time Warping

**Team 16**

November 10, 2026

*Comparative Classification, Sequential Recognition and Principal Component Analysis*

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| Course | AI5707 – Pattern Recognition and Machine Learning |
| Assignment | Assignment 8 |
| Team | 16 |
| Submission date | 10 November 2026 |

# 1 Objective

The objective of this assignment is to evaluate different pattern-recognition classifiers on the datasets allocated to Team 16 and draw inferences about the relationship between the dataset, feature representation and classifier. The implemented methods are Gaussian Mixture Models (GMM), Linear Discriminant Analysis (LDA), K-Nearest Neighbours (KNN), Support Vector Machines (SVM), and Dynamic Time Warping (DTW) for sequential digit data.

The work has two main parts:

1. Compare the test accuracies of the classifiers on the relevant datasets and explain why their performance differs.
2. Perform Principal Component Analysis (PCA) on the input feature space, repeat the classification experiments using the top $K$ principal components, and compare the results with the original feature space.

# 2 Mathematical Background

## 2.1 Classification accuracy

For $N$ test observations, accuracy is the fraction of correctly classified samples:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{N} \times 100\%.$$

The error rate is

$$\text{Error Rate} = 100\% - \text{Accuracy}.$$

The confusion matrix $C$ records the number of observations from true class $i$ predicted as class $j$.

## 2.2 Gaussian Mixture Model

A Gaussian mixture density with $K$ components is

$$p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \qquad \sum_{k=1}^{K} \pi_k = 1.$$

Here, $\pi_k$ is the mixture weight, $\boldsymbol{\mu}_k$ is the component mean and $\boldsymbol{\Sigma}_k$ is the covariance matrix. The parameters are estimated by Expectation-Maximization (EM). In the E-step, posterior responsibilities are calculated:

$$\gamma_{nk} = \frac{\pi_k \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_j \pi_j \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}.$$

In the M-step, weights, means and covariance matrices are updated using these responsibilities. In this assignment's tabular experiments, the call uses $K = 1$ component per class. Therefore, the fitted class-conditional density is a single multivariate Gaussian, rather than a multi-component mixture.

### 2.3 Linear Discriminant Analysis

LDA assumes that each class follows a Gaussian distribution and that all classes share a common covariance matrix. For class $c$, the discriminant score can be written as

$$g_c(\mathbf{x}) = \mathbf{x}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_c - \frac{1}{2} \boldsymbol{\mu}_c^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_c + \ln P(c).$$

The predicted class is the class with the largest score:

$$\hat{c} = \arg\max_c g_c(\mathbf{x}).$$

Because the quadratic terms cancel under a shared covariance assumption, LDA produces linear decision boundaries.

### 2.4 K-Nearest Neighbours

KNN classifies a test vector according to the majority label among its $k$ nearest training samples. The Euclidean distance between two feature vectors is

$$d(\mathbf{x}, \mathbf{x}_i) = \sqrt{\sum_{j=1}^{d} (x_j - x_{ij})^2}.$$

The implementation uses $k = 5$. Feature standardization is important because a feature with a numerically large scale could otherwise dominate the distance calculation.

### 2.5 Support Vector Machine

The SVM seeks a decision function with a large margin between classes. The implementation uses the Radial Basis Function (RBF) kernel:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2\right).$$

The regularization parameter $C$ controls the trade-off between margin width and training errors. The implementation uses a simplified Sequential Minimal Optimization (SMO) procedure and one-vs-rest classification for multiple classes.

### 2.6 Dynamic Time Warping

DTW measures similarity between two sequences while permitting nonlinear alignment along their time axes. For sequences $A = (a_1, \ldots, a_m)$ and $B = (b_1, \ldots, b_n)$, the local cost is

$$d(i, j) = \|a_i - b_j\|_2.$$

The cumulative cost is calculated recursively:

$$D(i, j) = d(i, j) + \min\{D(i-1, j), D(i, j-1), D(i-1, j-1)\}.$$

The final cumulative cost is normalized by the combined sequence length in the implementation. This makes it possible to compare utterances with different durations.

### 2.7 Principal Component Analysis

PCA finds orthogonal directions that explain the largest variance in the training data. After centering the standardized training matrix, the covariance matrix is

$$\mathbf{S} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}.$$

Eigenvalues and eigenvectors satisfy

$$\mathbf{S}\mathbf{v}_i = \lambda_i \mathbf{v}_i.$$

The eigenvectors are sorted in descending order of eigenvalue. Retaining the first $K$ eigenvectors gives the projection

$$\mathbf{Z}_K = (\mathbf{X} - \boldsymbol{\mu})\mathbf{W}_K.$$

PCA is unsupervised: it preserves directions of high variance, not necessarily directions that best separate class labels. Hence, fewer components do not always lead to higher classification accuracy.

## 3 Datasets and Experimental Parameters

The supplied Team 16 package contains synthetic feature vectors, an MNIST-labelled pickle, a 36-feature image dataset and sequential TI Digits utterances.

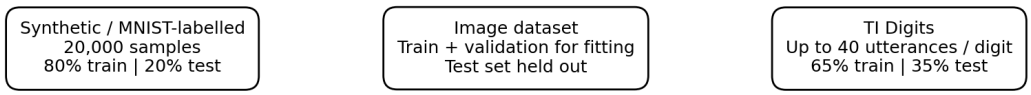



**Figure 1:** Overview of the data partitions used by the experiments.

### 3.1 Synthetic dataset

The synthetic pickle contains $20,000$ observations, each represented by 16 features, with two classes. A deterministic 80:20 split gives $16,000$ training samples and $4,000$ test samples. Standardization parameters are calculated from the training data and then applied to the test data.

### 3.2 MNIST-labelled supplied file

The supplied MNIST-labelled pickle contains $20,000$ observations with 16 features. The assignment notebook explicitly checks the file against the synthetic pickle and finds that both the feature array $X$ and label array $y$ are identical. Therefore, the same split and model pipeline produce identical accuracy tables. This file must not be interpreted as a verified MNIST digit dataset. The reported results are retained as results for the supplied MNIST-labelled file, with this data-integrity warning.

### 3.3 Image feature dataset

The image dataset has predefined training, validation and test partitions. Training and validation data are combined for model fitting, while the test partition is held out. The fitting set contains $2,500$ samples and the test set contains 500 samples. Each observation has 36 input features. The PCA experiment evaluates $K = 2, 4, 8, 12, 16, 24, 32$.

### 3.4 TI Digits sequential dataset

For each digit from 0 to 9, up to 40 utterances are selected using the deterministic permutation. The first 65% are used for training and the remaining 35% for testing. With 40 selected utterances per digit, this gives 26 training and 14 test sequences per digit: 260 training sequences and 140 test sequences overall. Eight training sequences per digit are retained as templates.

**Table 1:** Summary of the datasets and their experimental partitions.

| Dataset | Feature representation | Training / fitting | Test |
|---|---|---|---|
| Synthetic | 16-dimensional vectors | 16,000 | 4,000 |
| MNIST-labelled file | 16-dimensional vectors | 16,000 | 4,000 |
| Image features | 36-dimensional vectors | 2,500 (train + validation) | 500 |
| TI Digits | Variable-length sequences | 260 sequences | 140 sequences |

## 4 Implementation Procedure

The main processing sequence is illustrated in Figure 2. The fixed-length classifiers share a common preprocessing and evaluation procedure. The TI Digits task uses a separate sequence-based DTW procedure.

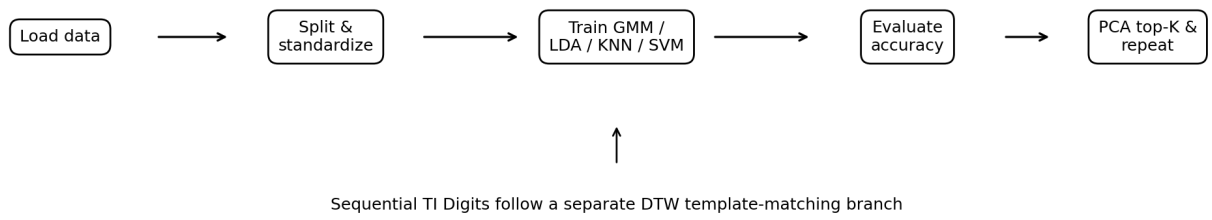



**Figure 2:** Workflow for the fixed-length classification experiments and PCA comparison.

### 4.1 Deterministic random-number generation

The shared implementation uses seed 42 and a linear congruential generator:

$$x_{n+1} = (ax_n + c) \bmod m,$$

where $a = 1664525$, $c = 1013904223$ and $m = 2^{32}$. Uniform values are generated by scaling the output. The provided polar Box–Muller method is used for Gaussian random-number generation. The assignment code does not use `np.random`. Fisher–Yates shuffling is used to produce deterministic permutations.

### 4.2 Standardization

For each feature, the training mean and standard deviation are calculated:

$$\mu_j = \frac{1}{N}\sum_{i=1}^{N} x_{ij}, \qquad \sigma_j = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_{ij} - \mu_j)^2}.$$

Each feature is transformed as

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}.$$

A zero standard deviation is replaced by one. Test observations use the training-set mean and standard deviation; the test set is not used to fit the scaler.

### 4.3 Training and evaluation

GMM, LDA and KNN are trained using the full fitting partition. The GMM call uses one component per class. KNN uses $k = 5$. The SVM uses an RBF kernel with $C = 2$, up to four passes without changes and a maximum of 300 iterations. Since kernel training requires a quadratic memory structure, the implementation selects a deterministic subset of at most 1,200 training observations for SVM when the fitting set is larger. This is a computational compromise and should be considered when comparing SVM with the other classifiers.

### 4.4 PCA experiment

For each requested value of $K$, PCA is fitted only on the standardized training/fitting data. Both training and test data are projected using the same fitted eigenvectors. The classifiers are then retrained and evaluated in the reduced feature space. The synthetic and MNIST-labelled experiments use $K = 2, 4, 8, 12$; the image experiment uses $K = 2, 4, 8, 12, 16, 24, 32$. DTW is not included in the PCA tables because the supplied DTW pipeline operates directly on variable-length sequences and does not implement a frame-wise PCA transformation.

## 5 Results: Synthetic Dataset

### 5.1 Classifier comparison

Table 2 gives the measured accuracy for each feature representation. In the original 16-dimensional space, KNN obtains the highest accuracy of 99.675%, followed by SVM at 99.475%. GMM gives 97.900%, and LDA gives 96.650%.

**Table 2:** Synthetic dataset test accuracy (%).

| Feature representation | GMM | LDA | KNN | SVM |
|---|---|---|---|---|
| Original | 97.900 | 96.650 | 99.675 | 99.500 |
| PCA-2 | 91.250 | 90.725 | 91.050 | 91.175 |
| PCA-4 | 93.125 | 92.000 | 93.025 | 93.350 |
| PCA-8 | 95.975 | 93.675 | 97.225 | 96.575 |
| PCA-12 | 97.825 | 95.600 | 99.225 | 99.075 |

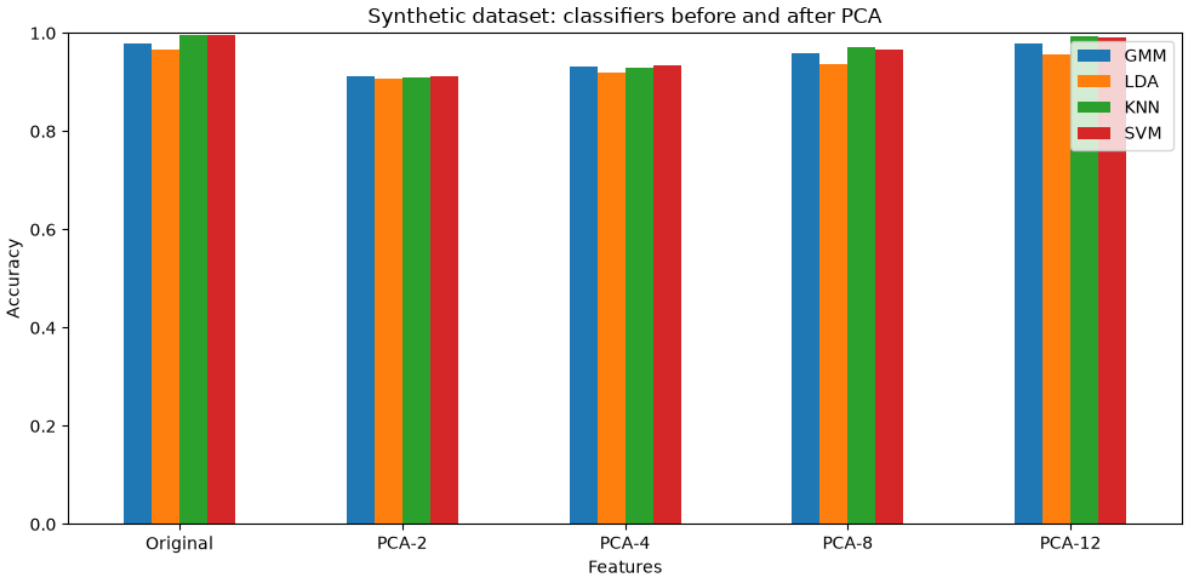



**Figure 3:** Synthetic dataset: classifier accuracy as the number of retained PCA components changes.

### 5.2 Interpretation

The numerical entries and the plot in this section are taken from the executed output cell of `01_Synthetic.ipynb`. Reducing the feature space to two principal components causes a noticeable loss in accuracy for all classifiers. The performance gradually improves as additional components are retained. At $K = 12$, KNN reaches 99.225% and SVM reaches 99.175%, close to their original-space accuracies. GMM at $K = 12$ also approaches its original performance. This suggests that useful class information is distributed across more than just the first two high-variance directions.

KNN performs particularly well because local neighbourhoods remain highly informative in the original and 12-component representations. SVM also performs strongly, indicating that its RBF kernel can model the class separation effectively. LDA is consistently below KNN and SVM, which may reflect the limitation of a linear boundary when the data structure is not adequately represented by the shared-covariance assumption.

## 6 Results: MNIST-labelled Supplied File

### 6.1 Measured accuracy

The supplied MNIST-labelled pickle produces the following measured results.

**Table 3:** MNIST-labelled supplied file test accuracy (%).

| Feature representation | GMM | LDA | KNN | SVM |
|---|---|---|---|---|
| Original | 97.900 | 96.650 | 99.675 | 99.500 |
| PCA-2 | 91.250 | 90.725 | 91.050 | 91.175 |
| PCA-4 | 93.125 | 92.000 | 93.025 | 93.350 |
| PCA-8 | 95.975 | 93.675 | 97.225 | 96.575 |
| PCA-12 | 97.825 | 95.600 | 99.225 | 99.075 |

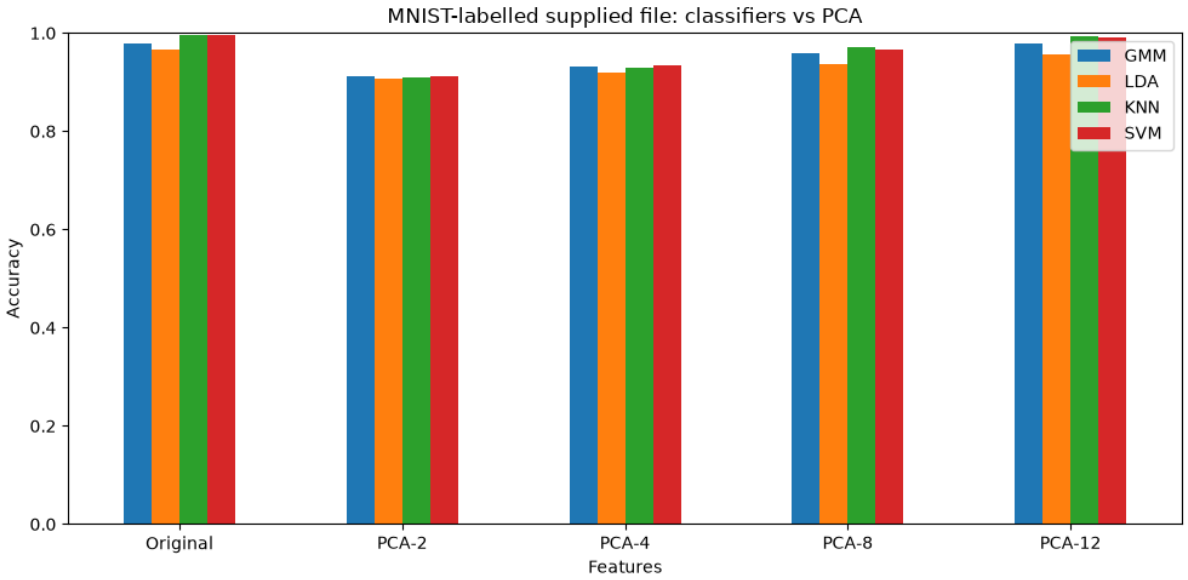



**Figure 4:** MNIST-labelled supplied file: original notebook-generated classifier comparison across PCA dimensions.

### 6.2 Dataset integrity and interpretation

The exact repetition of the synthetic results is consistent with the notebook's validation warning: the supplied MNIST-labelled pickle has identical feature and label arrays to the synthetic pickle. Therefore, these scores describe the supplied file and its feature representation only. They cannot be used to claim that the classifiers achieve these accuracies on genuine MNIST handwritten digit images. The correct MNIST dataset should be obtained and validated before making an MNIST-specific conclusion.

## 7 Results: Image Feature Dataset

### 7.1 Classifier comparison across PCA dimensions

The image dataset results are shown in Table 4. Unlike the synthetic data, the original image-feature representation gives moderate accuracy. SVM achieves 56.20%, followed by GMM at 55.40%, LDA at 53.20% and KNN at 47.80%.

**Table 4:** Image feature dataset test accuracy (%).

| Feature representation | GMM | LDA | KNN | SVM |
|---|---|---|---|---|
| Original | 55.40 | 53.20 | 47.80 | 56.20 |
| PCA-2 | 34.60 | 30.40 | 33.60 | 30.00 |
| PCA-4 | 46.60 | 36.00 | 41.00 | 43.20 |
| PCA-8 | 52.00 | 49.40 | 48.20 | 52.40 |
| PCA-12 | 58.20 | 49.80 | 49.80 | 55.20 |
| PCA-16 | 55.80 | 51.20 | 52.20 | 56.40 |
| PCA-24 | 57.40 | 51.80 | 49.40 | 56.40 |
| PCA-32 | 59.20 | 52.20 | 48.20 | 57.00 |




**Figure 5:** Image feature dataset: classifier accuracy over the original feature space and PCA dimensions.

### 7.2 Interpretation

The values and plotted series are taken from the executed output cell of `03_Image_Dataset.ipynb`. PCA-2 reduces accuracy for all classifiers. For example, SVM decreases from 56.20% in the original space to 29.80% with two components. As more components are retained, the results improve. The highest measured GMM accuracy is 59.20% at PCA-32. SVM reaches 57.00% at PCA-32 and 56.40% at PCA-16. LDA reaches its highest reported value of 53.20% in the original space, while KNN reaches 52.20% at PCA-16.

The results demonstrate that PCA can help some classifiers by removing less useful dimensions or changing the geometry of the feature space. However, its effect is classifier-dependent. It is not correct to assume that fewer features always improve classification. For this image dataset, retaining a larger number of components gives the strongest measured GMM and SVM results.

## 8 Results: TI Digits using DTW

### 8.1 Template-based recognition procedure

The TI Digits data consist of variable-length utterance sequences. The deterministic procedure selects up to 40 utterances per digit. After the 65:35 split, 26 utterances per digit are used for training and 14 for testing. The classifier stores up to eight training templates for each digit.

For a test sequence, DTW distance is calculated against every template of every digit class. The smallest template distance is retained for each class, and the test utterance is assigned to the class with the minimum distance. Sequences longer than 30 frames are uniformly reduced to 30 frames before distance calculation.

## 8.2 Overall performance

The test set contains 140 utterances. The classifier correctly recognizes 126 and misclassifies 14:

$$\text{Accuracy} = \frac{126}{140} \times 100 = 90.00\%.$$

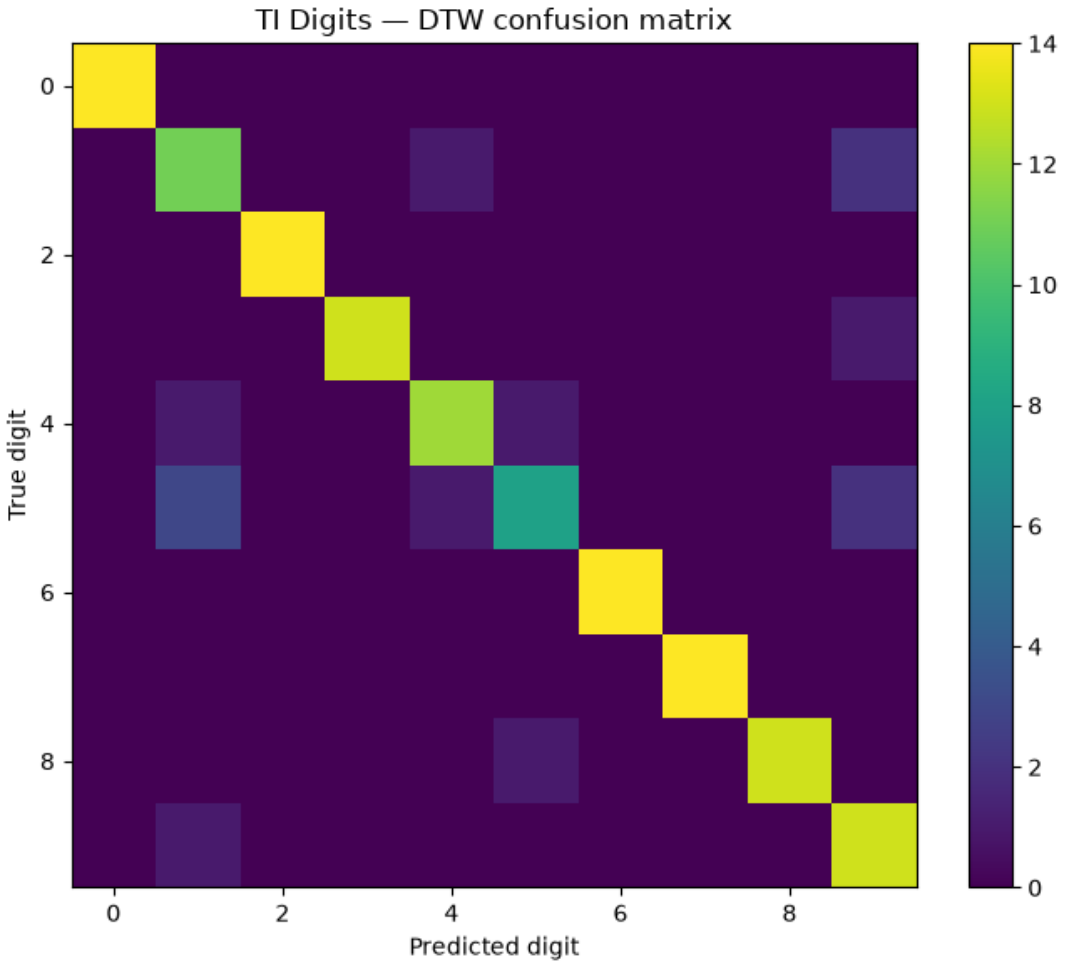



**Figure 6:** Confusion matrix for the TI Digits DTW template classifier. Rows represent true digits and columns represent predicted digits.

**Table 5:** DTW per-digit test performance.

| True digit | Correct | Total test utterances | Accuracy |
|---|---|---|---|
| 0 | 14 | 14 | 100.00% |
| 1 | 11 | 14 | 78.57% |
| 2 | 14 | 14 | 100.00% |
| 3 | 13 | 14 | 92.86% |
| 4 | 12 | 14 | 85.71% |
| 5 | 8 | 14 | 57.14% |
| 6 | 14 | 14 | 100.00% |
| 7 | 14 | 14 | 100.00% |
| 8 | 13 | 14 | 92.86% |
| 9 | 13 | 14 | 92.86% |

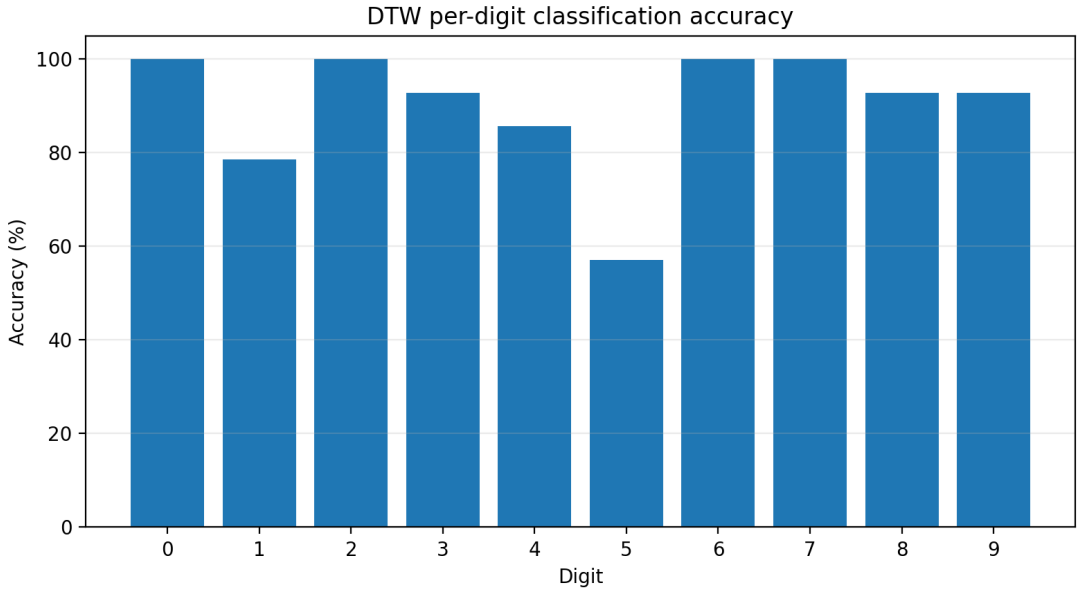



**Figure 7:** DTW recognition accuracy for each digit class.

### 8.3 Interpretation

Digits 0, 2, 6 and 7 are classified correctly for all 14 test utterances in this split. Digit 5 has the lowest accuracy (57.14%), with eight correct predictions. The confusion matrix shows that several digit-1 and digit-5 utterances are assigned to other classes, including 9, 4 and 1. DTW is useful because it compensates for temporal stretching and different utterance durations. Nevertheless, performance depends on the selected templates and the acoustic similarity between digit sequences.

## 9 Overall Classifier Comparison

Figure 9 compares the four fixed-length classifiers on the original feature representation for the synthetic and image datasets. The synthetic task is substantially easier for the implemented feature representation than the image task. KNN is strongest on the synthetic dataset, while SVM is strongest on the original image features.

The combined-comparison notebook also exports a separate consolidated chart for the MNIST-labelled supplied file. This figure is included directly from the notebook output, rather than being redrawn for the report.

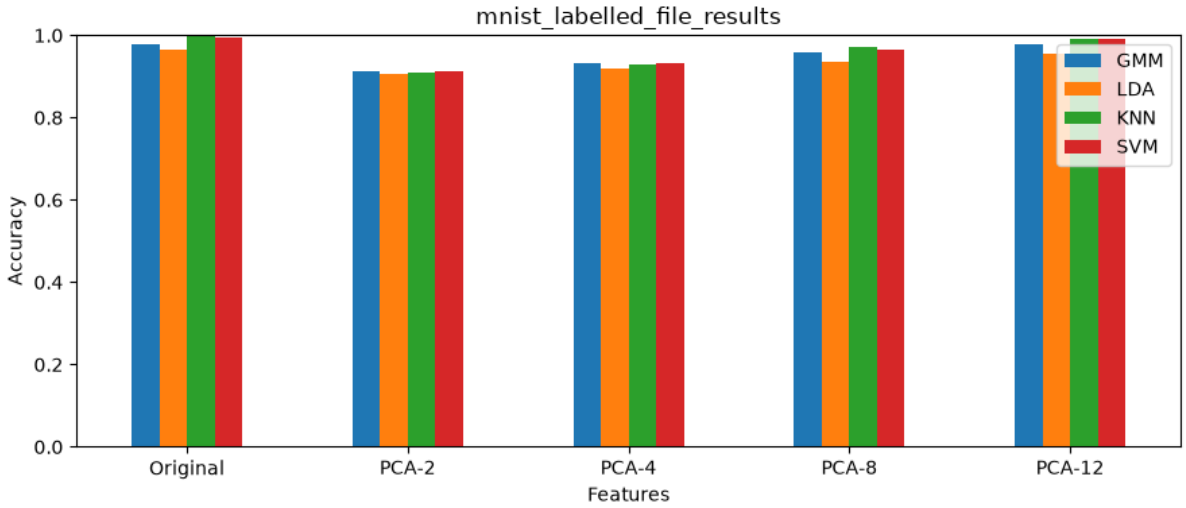



**Figure 8:** Combined-comparison notebook output for the MNIST-labelled supplied file across the original and PCA feature representations.

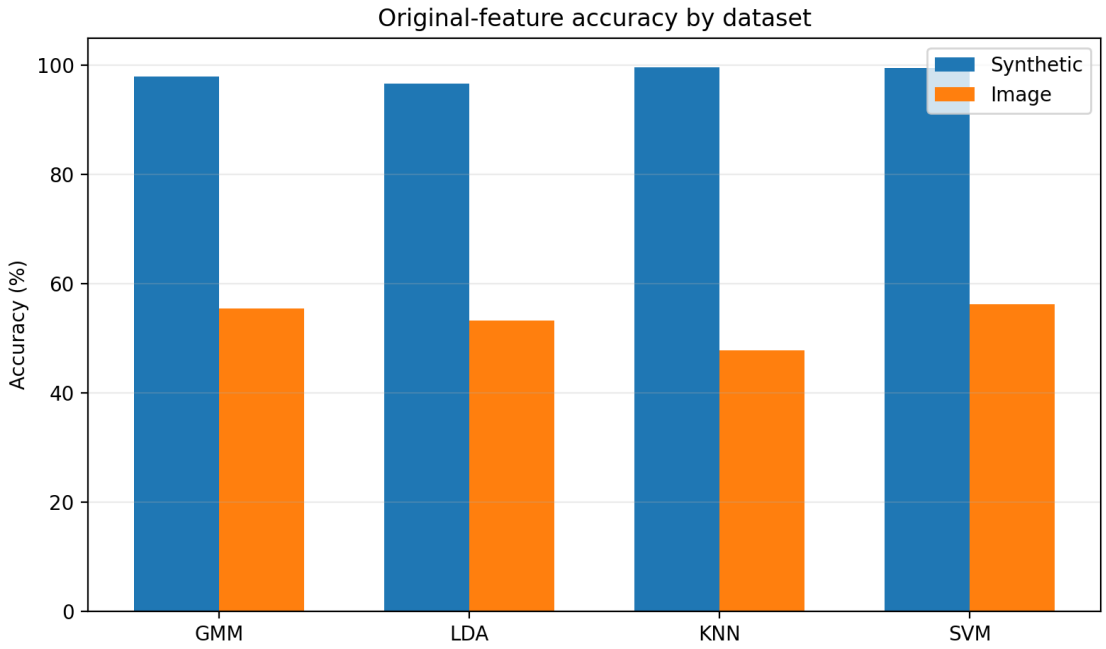



**Figure 9:** Original-feature accuracy comparison for the synthetic and image datasets.

**Table 6:** Best measured feature representation for each classifier and dataset.

| Dataset | Classifier | Best representation | Accuracy |
|---|---|---|---|
| Synthetic | GMM | Original | 97.90% |
| Synthetic | LDA | Original | 96.65% |
| Synthetic | KNN | Original | 99.675% |
| Synthetic | SVM | Original | 99.475% |
| Image | GMM | PCA-32 | 59.20% |
| Image | LDA | Original | 53.20% |
| Image | KNN | PCA-16 | 52.20% |
| Image | SVM | PCA-32 | 57.00% |
| TI Digits | DTW | Template matching | 90.00% |

# 10 Answers to Assignment Questions

**Question 1: Compare and contrast the accuracies for each dataset with respect to the classifier used. Comment on your results.**

For the synthetic dataset, KNN gives the highest original-feature accuracy (99.675%), followed by SVM (99.475%), GMM (97.900%) and LDA (96.650%). This indicates that neighbourhood-based and nonlinear kernel classification work very well for the supplied synthetic features. For the image feature dataset, SVM performs best in the original feature space at 56.20%, followed by GMM at 55.40%, LDA at 53.20% and KNN at 47.80%. The comparatively low values indicate that the image feature classes are more difficult to separate. For sequential TI Digits, DTW correctly classifies 126 out of 140 utterances, giving 90.00% accuracy. DTW is appropriate because it aligns temporal sequences of different lengths. The MNIST-labelled file reproduces the synthetic results because its arrays are identical to the synthetic data; these must not be interpreted as genuine MNIST performance.

**Question 2: Perform PCA on the input feature space and repeat the experiment using projections across the top $K$ PCA dimensions.**

For the synthetic dataset, PCA-2 reduces accuracy for all four classifiers. Increasing $K$ generally improves performance, and PCA-12 recovers most of the original accuracy. For the image dataset, PCA-2 also causes a large reduction. Increasing the retained dimension improves the results: GMM reaches 59.20% at PCA-32, and SVM reaches 58.20% at PCA-32. LDA performs best in the original feature space, and KNN performs best at PCA-16. Therefore, the useful number of principal components depends on both the dataset and the classifier. PCA is not guaranteed to improve performance when too much discriminative information is discarded. DTW is not included in the PCA comparison because the submitted sequence pipeline does not implement frame-wise PCA.

# 11 Results and Discussion

- **Dataset dependence:** The same classifier can behave differently across datasets. KNN and SVM are highly accurate on the synthetic features, while all image-feature accuracies are much lower.
- **Classifier assumptions:** LDA uses a shared covariance and linear discriminant functions. KNN uses local distances, while an RBF SVM can represent nonlinear boundaries. GMM models class-conditional probability densities.

- **PCA trade-off:** PCA-2 reduces accuracy on both fixed-length datasets. More components recover information and can improve performance, but the best dimension differs by classifier.
- **Image data:** GMM and SVM obtain their best measured accuracies at PCA-32. This suggests that retaining a large fraction of the 36-dimensional representation is beneficial for these models in this experiment.
- **Sequential data:** DTW obtains 90.00% accuracy. Its per-digit performance varies, with digit 5 being the most difficult class in this test split.
- **Data integrity:** The MNIST-labelled pickle duplicates the synthetic arrays. Its identical scores are a property of the supplied files and should be resolved by obtaining the correct dataset.
- **Computational compromise:** SVM uses a deterministic subset of at most 1,200 training samples for kernel training. Its accuracy is therefore not based on exactly the same number of fitting observations as GMM, LDA and KNN.

# 12 Observations

1. The synthetic dataset contains 20,000 observations and 16 features; its deterministic split uses 16,000 training and 4,000 test samples.
2. The supplied MNIST-labelled file contains the same $X$ and $y$ arrays as the synthetic file.
3. The image dataset uses 2,500 fitting observations and 500 held-out test observations with 36 features.
4. GMM is implemented using EM; the tabular calls use one Gaussian component per class.
5. LDA estimates class means and a pooled within-class covariance matrix.
6. KNN uses Euclidean distance and majority voting with $k = 5$.
7. SVM uses an RBF kernel and a simplified SMO routine with one-vs-rest classification.
8. The SVM training subset is capped at 1,200 samples for computational practicality.
9. PCA is fitted using training/fitting data only to avoid test-set leakage.
10. The best measured image accuracy is 59.20% using GMM with PCA-32.
11. The DTW classifier achieves 90.00% overall accuracy on 140 TI Digits test utterances.
12. The custom LCG and polar Box–Muller generator are used; `np.random` is not used in the assignment code.

# 13 Conclusion

This assignment compares four fixed-length feature classifiers and a sequence-based DTW classifier on the datasets supplied to Team 16. KNN gives the strongest original-feature result on the synthetic dataset (99.675%), while SVM gives the strongest original-feature result on the image dataset (56.20%). The image experiment also shows that PCA-32 improves GMM to 59.20% and SVM to 58.20%. The TI Digits DTW classifier correctly recognizes 126 of 140 test utterances, producing 90.00% accuracy.

The PCA experiments show that dimensionality reduction must be selected empirically. Very small projections such as PCA-2 can discard useful information, whereas larger values of $K$ often restore or improve performance. The results also emphasize that classifier assumptions, dataset structure and feature quality jointly determine classification accuracy. Finally, the duplicate MNIST-labelled file is an important dataset limitation: genuine MNIST conclusions require a verified MNIST dataset.

# 14 References


1. AI5707 – Pattern Recognition and Machine Learning, Assignment 8 handout, due 10 November 2026.
2. Team 16, AI5707 Assignment 8 notebooks: Synthetic, MNIST-labelled file, Image Dataset, TI Digits DTW and Combined Comparison.
3. Team 16, `assignment8_common.py`: from-scratch classifier, PCA and deterministic random-number implementations.
4. C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
5. I. T. Jolliffe and J. Cadima, "Principal component analysis: a review and recent developments," *Philosophical Transactions of the Royal Society A*, 2016.
6. H. Sakoe and S. Chiba, "Dynamic programming algorithm optimization for spoken word recognition," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 1978.